Baseline Toy VELO Reconstruction Report

Derived from `baseline.ipynb`

Quantum Track Reconstruction

April 2, 2026

Abstract

This report documents the workflow and observed outputs in `baseline.ipynb` in `Toy_Characterisation`. The notebook establishes a clean baseline for toy VELO track reconstruction, validates reconstruction quality on single and multi-event samples, and studies pairwise segment-angle separation between true and false combinations.

1 Objective

The baseline notebook is organized into four tasks:

1. Build detector geometry and event-generation helpers.
2. Reconstruct a clean 3-track event with a Hamiltonian solver and verify truth matching.
3. Run a 10-event mixed-occupancy check and summarise efficiency/ghost behaviour.
4. Measure true-vs-false pairwise segment-angle distributions, including a large-sample study (100 events $\times$ 50 tracks).

2 Setup and Definitions

2.1 Geometry and Event Generation

The baseline geometry and generation settings are:

- 5 modules with positions $z = [100, 133, 166, 199, 232]$ mm ($\Delta z = 33$ mm).
- Module half-widths $l_x = l_y = 50$ mm.
- Primary-vertex and track generation via `StateEventGenerator`.

2.2 Acceptance Scale

The segment-angle acceptance threshold is computed as

$$\varepsilon = s\sqrt{2\theta_s^2 + 12\theta_r^2 + 2\theta_{\min}^2}, \tag{1}$$

with

$$\theta_s = \sigma_{\text{scatt}}, \qquad \theta_r = \arctan\left(\frac{\sigma_{\text{res}}}{\Delta z}\right), \tag{2}$$

and scale factor $s = 3$ in this notebook.

For the clean baseline:

- $\sigma_{\text{res}} = 0$,

- $\sigma_{\text{scatt}} = 10^{-3}$ rad,
- $\Delta z = 33$ mm,

which yields

$$\varepsilon = 0.004243 \text{ rad} = 4.243 \text{ mrad}. \tag{3}$$

3 Hamiltonian Reconstruction Baseline

The clean event contains 3 truth tracks and 15 hits (5 hits per track). Hamiltonian settings are $\gamma = 1.5$, $\delta = 1.0$, and ε above.

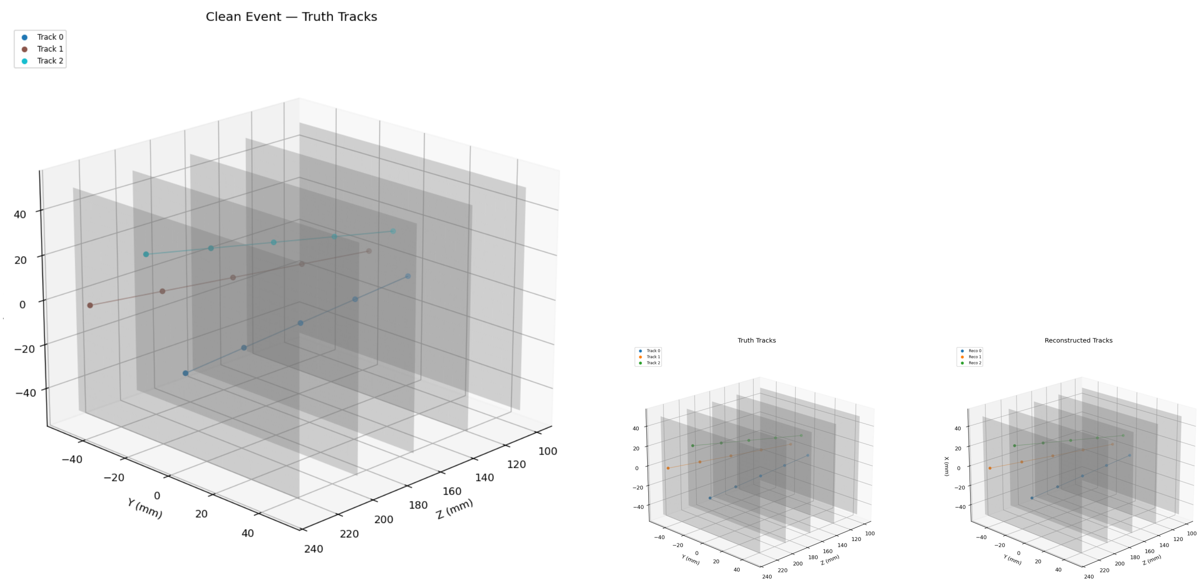
The notebook uses

$$x_{\text{baseline}} = \frac{\delta}{\delta + \gamma} = 0.4, \qquad x_{\text{thr}} = \frac{1 + x_{\text{baseline}}}{2} = 0.7. \tag{4}$$

Observed solver output:

- Number of segments: 36.
- Hamiltonian matrix: 36×36, nnz $= 54$.
- Active above threshold: 12/36.
- Activation range: $[0.4000, 1.2727]$.

Reconstruction returns 3 tracks, each with 5 hits.



(a) 3-D truth event display (3 tracks).

(b) Reconstructed vs. truth tracks.

Figure 1: Clean 3-track event. Left: truth geometry. Right: reconstructed overlay.

4 Clean-Event Validation

Using `EventValidator` with `purity_min=0.7`, all three truth tracks are recovered as primary matches with purity and hit efficiency equal to 1.0.

Table 1: Clean-event summary metrics.

Metric	Value
Efficiency	1.0
Ghost rate	0.0
Clone fraction	0.0
Mean purity	1.0
Hit efficiency	1.0
Matched / reconstructible	3 / 3

For consecutive-segment pair angles on the clean event:

- True pairs: 9, mean angle 0.000000 rad.
- Reconstructed pairs: 9, mean angle 0.000000 rad.
- Reference threshold: $\varepsilon = 0.004243$ rad.

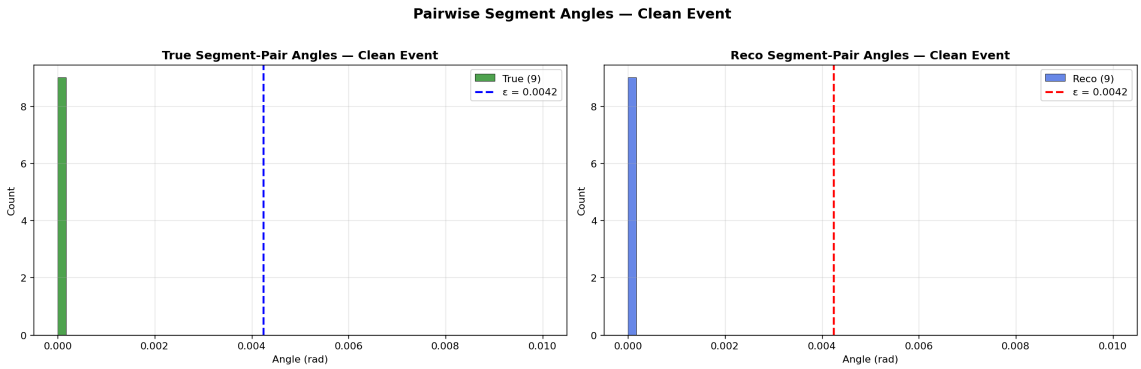


Figure 2: Pairwise segment-pair angles on the clean 3-track event. Left: true pairs (green). Right: reconstructed pairs (blue). All angles lie at exactly zero because the straight-line toy tracks have no multiple scattering.

5 10-Event Mixed-Occupancy Study

Track-count schedule: $[2, 3, 4, 5, 6, 3, 4, 5, 2, 4]$.

Table 2: Per-event baseline summary from notebook output.

Event	True	Reco	Eff	Ghost	TrueSeg	RecoSeg
0	2	2	1.000	0.000	8	8
1	3	3	1.000	0.000	12	12
2	4	4	1.000	0.000	16	16
3	5	5	1.000	0.000	20	20
4	6	2	0.333	0.000	16	8
5	3	3	1.000	0.000	12	12
6	4	3	0.500	0.333	16	17
7	5	5	1.000	0.000	20	20
8	2	2	1.000	0.000	8	8
9	4	4	1.000	0.000	16	16

Aggregate over the 10 events:

- Mean efficiency: 0.883.
- Mean ghost rate: 0.033.
- Most events are perfectly reconstructed; notable degradation appears in events 4 and 6.

6 Pairwise Segment-Angle Separation

Angles are computed between segment pairs sharing a middle hit:

$$\alpha = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \, \|\mathbf{v}_2\|}\right), \tag{5}$$

where $\mathbf{v}_1$ and $\mathbf{v}_2$ are incoming/outgoing segment vectors. A pair is labeled true only if all three hits are on the same truth track.

Combined over the 10-event sample:

- 106 true pairs,
- 1,742 false pairs.

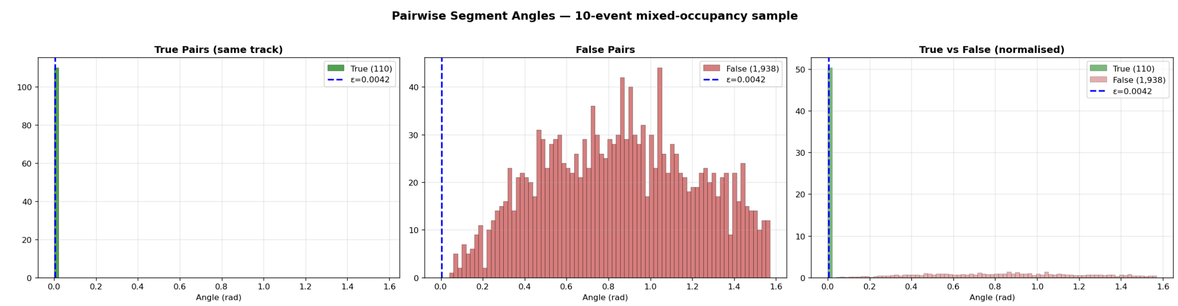


Figure 3: Pairwise segment-pair angles aggregated over the 10-event mixed-occupancy sample. Left: true pairs; centre: false pairs; right: normalised overlay. True pairs cluster near zero; false pairs are broadly distributed, confirming the threshold ε cleanly separates the two classes.

Large-sample study (20 events $\times$ 50 tracks):

- 2,975 true segment pairs,
- 7,430,425 false segment pairs.

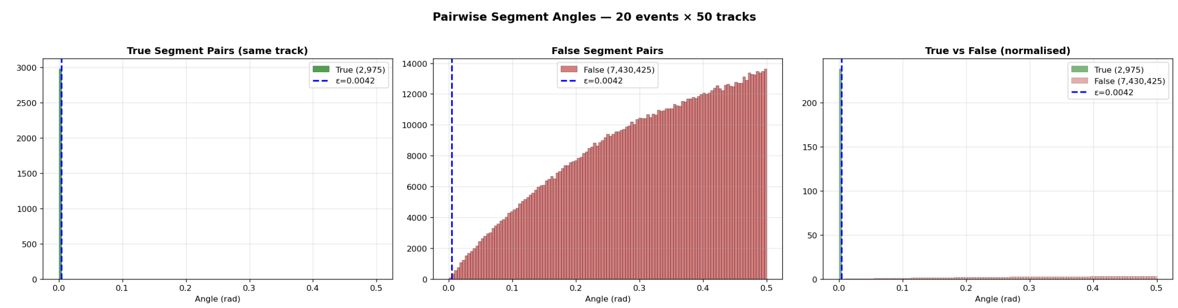


Figure 4: Pairwise segment-pair angles for the large-sample run (20 events $\times$ 50 tracks). The true-pair peak at zero and the broad false-pair distribution are consistent with the 10-event results, confirming stability at higher occupancy.

This large imbalance is expected combinatorially and motivates threshold-based filtering in downstream reconstruction logic.

7 Conclusion

The baseline notebook demonstrates that the Hamiltonian-based reconstruction is exact on the clean event and robust for most moderate-occupancy events. The angle-distribution studies provide the intended separation context for true vs false segment-pair combinations, and the 100×50 run establishes a high-statistics baseline for future tuning and comparisons.